Investigating the relationship between mutational spectra and antimicrobial-resistance burden in MRSA

Introduction

Methicillin-resistant *Staphylococcus aureus* (MRSA) remains a major public-health concern. Although the use of several antibiotic classes has declined in recent years, the evolutionary dynamics of resistance in MRSA remain complex. Mutational spectra, which capture characteristic patterns of nucleotide substitutions, reflect underlying biological processes such as oxidative stress, DNA repair activity, or selective pressures. As such, they provide a useful framework for exploring how environmental and clinical factors shape the evolution of MRSA.

Aims

To model the relationship between mutational spectra and AMR burden, with and without cohort effects representing isolates collected in different time periods, using linear models and interaction terms.

Methods

I will use AMRFinderPlus to screen around 250 MRSA genomes for acquired antimicrobial-resistance genes. For each isolate, I will record the total number of resistance genes and the number of antibiotic classes represented. To obtain a recombination-free core genome alignment, I will run Gubbins and produce a concatenated multiple sequence alignment (MSA) representing vertically inherited SNPs. Using the recombination-filtered MSA, I will reconstruct maximum-likelihood phylogenies with RAxML. The resulting phylogenetic trees will serve as the basis for downstream mutational-spectrum analyses. I will compute mutational spectra with MutTui, for each terminal branch, I will extract a per-isolate mutational spectrum. I will combine mutational spectra across tip branches with the same AMR burden and then perform principal component analysis (PCA) on the combined mutational spectra. To assess the relationship between mutational spectra and antimicrobial-resistance burden, I will fit linear models.

Importance

The planned analyses integrate multiple evolutionary, genomic, and epidemiological perspectives to understand how *S. aureus* evolves under antimicrobial pressure. This is essential for predicting future resistance trends and the dynamics of bacterial populations.

References

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